

A Simulation-Based Machine Learning Framework for Quantum Noise-Channel Classification

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Abstract—Quantum teleportation (QT) is the most essential component of quantum communication, but its performance falls significantly when noisy intermediate-scale quantum (NISQ) channels are present. This paper presents a physics-informed machine learning (ML) framework for classifying quantum noise channels in teleportation systems. We employ Kraus operators to model five representative noise processes: depolarizing, amplitude damping, phase damping, bit flip, and phase flip. Additionally, we simulate their effects on teleportation fidelity and entanglement. A dataset comprising 15,360 instances is generated using the Monte Carlo method, featuring 35-dimensional vectors derived from quantum information metrics, including fidelity, entropy, purity, and Quantum Fisher Information (QFI). Multiple classifiers are evaluated, including Support Vector Machines (SVM), Random Forest (RF), Multilayer Perceptron (MLP), and Gradient Boosting (GB). The results indicate that ensemble methods demonstrate superior performance compared to alternative models. Specifically, GB attained the highest accuracy of 96.5%, ensuring strong per-class performance across all categories of errors. These findings highlight the potential of physics-informed ML for developing noise-aware and reliable quantum communication systems.

Index Terms—Quantum teleportation, Noise, Machine learning, Classification.

I. INTRODUCTION

Quantum teleportation is a fundamental mechanism in quantum communication that allows quantum states to be transferred between distant nodes using entanglement and classical communication. Quantum teleportation, first introduced by [1], uses a shared entangled pair and classical communication to transfer an unknown quantum state (e.g., Bell state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$). However, in ideal circumstances, it provides complete fidelity, but its actual application in quantum networks is hindered by noisy intermediate-scale quantum (NISQ) devices that exhibit decoherence, gate faults, and channel noise [2]. While photon loss in free space or fiber-optic channels restricts long-distance teleportation, decoherence, which is brought on by environmental interactions, weakens entanglement [3]. For instance, studies conducted beyond 100 km have observed fidelity losses due to air turbulence and absorption [4]. Those challenges degrade entanglement, decreasing teleportation fidelity and cutting off scalability. Recent research suggests new methods for noise-resistant teleportation that provide optimal fidelity without

the need for more qubits by utilizing generalized quantum measurements and control units (such as rotation gates and CNOT). But these approaches aren't scalable for big networks and have limited resources. Quantum network scalability is significantly hampered by noisy quantum channels, which are defined by amplitude damping, phase damping, or depolarizing noise. The no-cloning theorem inhibits direct signal amplification, requiring alternate methods such as quantum repeaters, which employ entanglement switching to enhance range [5].

Artificial Intelligence(AI) has become a significant factor in the progression of quantum communication [6]. AI-driven techniques, specifically ML-based models including Neural Networks(NN), Reinforcement Learning (RL), and SVM, are being employed to tackle intricate quantum challenges [7], [8]. ML is utilized in various aspects of quantum communication, such as error correction, noise reduction, optimization of quantum channels, and distribution of entanglement [9]. Furthermore, AI facilitates the acceleration of quantum key distribution, improves protocol security, and optimizes resource management within extensive quantum networks. One of the major challenges in quantum communication is its susceptibility to environmental disturbances, including photon loss, decoherence, and quantum noise, which compromise signal integrity [10]. AI algorithms are essential for the detection, prediction, and mitigation of errors in real time, thereby enhancing the reliability and efficiency of quantum communication networks [11]. Additionally, AI enhances adaptive quantum repeater networks, thereby increasing the range of QKD for secure communication over long distances [12]. As quantum technologies grow, ML will be a crucial tool for creating fault-tolerant quantum networks, refining quantum protocols, and guaranteeing safe, effective quantum cryptography systems.

This research presents a framework for noisy quantum channel classification in teleportation networks. The major contribution of this research can be summarized as follows:

- We developed a physics-informed machine learning framework that integrates Kraus-based quantum noise modeling with quantum teleportation simulations to classify five major quantum noise channels.
- This work introduces a comprehensive simulation dataset containing 15,360 quantum channel instances represented

by 35-dimensional feature vectors derived from quantum information metrics, including fidelity, entropy, purity, and Quantum Fisher Information.

- The proposed framework evaluates multiple machine learning classifiers and demonstrates that Gradient Boosting achieves superior classification performance with 96.48% accuracy, enabling efficient noise identification for reliable and scalable quantum communication systems.

The remaining portions of the document are arranged as follows: In Section 2, we survey relevant literature on noisy QT channel classification, and in Section 3, we lay out the methods in detail. In Section 4, we elucidate the experimental outcomes. The work is concluded in Section 5, which details the limitations and offers suggestions for future research.

II. REALTED WORK

In today's world, quantum systems can be precisely controlled to carry out new kinds of operations by taking advantage of their coherence features, like entanglement and superposition. But ambient noise causes quantum operations to lose precision and decoherence; hence, quantum behavior disappears. The paper [13] presents a Gaussian Mixture Model-Expectation Maximization(GMM-EM) method for modeling hybrid quantum noise (Poisson shot noise + AWGN) in Gaussian quantum channels. The approach captures overlapping, variable-weighted noise distributions more accurately than K-means or DBSCAN by lowering and optimizing Gaussian components. Simulations on synthetic bivariate HQN data reveal that GMM-EM produces improved clustering metrics (AIC, BIC, silhouette) and the greatest possible rates for both general and satellite quantum channels, with significant increases at 25 dB SNR. Another study [14] examines Bell, GHZ, and W states in the context of phase damping, amplitude damping, and depolarizing noise. It demonstrates that Bell states exhibit the highest fidelity in certain damping scenarios, whereas *GHZ* and *W* states display greater resilience in the presence of multi-qubit noise. The findings provide insights for the selection of optimal channels in noisy quantum communication. The research [15] presents a hybrid classification model that integrates classical preprocessing with quantum circuits, addressing amplitude and phase damping noise as represented by Kraus operators. In the Iris dataset, features are encoded into quantum states, processed via dissipative two-qubit channels, then measured for classification purposes. The model attains an accuracy of up to 96% in low-noise environments, demonstrating enhanced resilience to amplitude damping and significant promise for NISQ applications.

In this paper [16], a noise-aware VQA framework trains parameterized circuits for variational ground-state preparation under depolarizing, amplitude, and phase damping noise. It uses circuit compilation to create device-specific hardware-efficient implementations. The approach delivers up to 96% accuracy on noisy simulators and increased hardware fidelity on classification and state preparation tasks, demonstrating the benefits of noise modeling and compilation optimization. Here,

the author [17] presents an effective technique for studying the model and detecting the noise of quantum channels using a quantum neural network(QNN). They develop the architecture of the QNN and the model training procedure based on different kinds of noisy quantum channel models. The cost function of the QNN may approach 90% of the channel model, according to the findings of our experiments, which we conducted to confirm the scheme's training efficacy. This research [18] presents a quantum teleportation error mitigation strategy utilizing quantum topological semion codes and a convolutional neural network (CNN) decoder to safeguard states against noise. The double-semion code is represented in a toric-code-like 8×8 configuration, enabling the CNN to forecast logical-level corrections inside a depolarizing noise framework. It is trained on diverse error rates, attaining thresholds of 0.082 (< 0.007 dB) and 0.096 (< 0.01 dB), exceeding most classical decoders. Another study suggests a technique for encoding qubits in two time intervals utilizing "time-bin entanglement" for long-distance quantum communication. This method protects qubits from noise without needing more resources as the channel length grows, which keeps the fidelity high. The setup, which includes optical parts like half-wave plates and polarizing beam splitters, can be tested and allows for safe, high-quality transmission [19].

III. METHODOLOGY

This section presents the specific steps to take in order to classify quantum channels in noisy quantum teleportation networks using a combination of ML and quantum information theory. By integrating theoretical and experimental factors, the method aims to facilitate the optimization of protocols in real-time. Modeling quantum channels, simulating the quantum teleportation protocol, extracting features using quantum information metrics, and classifying quantum channels using ML are the components that make up the methodology.

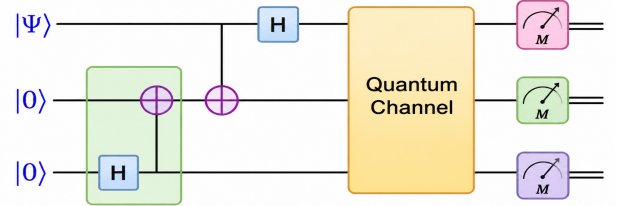


Fig. 1. Proposed workflow for noisy quantum teleportation channel classification.

The circuit in Figure 1 illustrates an approach for quantum communication, most likely for quantum teleportation. After the qubits in the $|0\rangle$ state are prepared, they are put through Hadamard and CNOT gates to generate entanglement between them. The entangled qubits are transmitted across the quantum channel, and the qubits undergo final measurements, denoted by M . Measurements cause the quantum state to collapse, which enables classical communication to finish teleporting

the state $|\psi\rangle$. Here we see the fundamental quantum communication phases of entanglement distribution, quantum state transmission, and measurement.

A. Quantum Channel Modeling and Noise Analysis

Modeling quantum channels is the first and most important part of our methodology. This is necessary for simulating realistic noise circumstances in quantum teleportation networks. Here, quantum channels are represented by Kraus operators, which stand for the development of quantum states subjected to different types of noise variations. We simulate several kinds of noise, each of which stands for a distinct physical mechanism that degrades the quantum state throughout transmission.

1) *Pauli Matrices and Quantum States*: A single qubit's quantum state is denoted by a density matrix *density matrix* ρ , defined as:

$$\rho = |\psi\rangle\langle\psi| \quad (1)$$

where $|\psi\rangle$ is a quantum state vector that has been normalized. When different Pauli matrices are used, the evolution of the quantum state changes. The bit-flip, phase-flip, and depolarizing noise are represented by the Pauli matrices X, Y, Z :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (2)$$

These matrices are essential for the modeling of noise and quantum error correction, as they characterize the primary quantum errors that manifest in communication channels.

2) *Kraus Operators for Quantum Channel Types*: We define various quantum channels that represent typical quantum teleportation noise sources. To characterize the quantum state evolution as a result of these noise processes, the *Kraus operators* for each channel are constructed. Here are some quantum channels that are being considered:

- **Depolarizing channel**: The depolarizing channel introduces stochastic errors and is modeled as follows:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X^\dagger + Y\rho Y^\dagger + Z\rho Z^\dagger) \quad (3)$$

where p is the probability of error, and X, Y, Z are the Pauli matrices.

- **Amplitude damping channel**: The amplitude damping channel simulates energy loss (T1 decay) and is characterized by the subsequent Kraus operators:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \quad (4)$$

Here, the damping parameter γ measures the amount of energy lost to the environment.

- **Phase damping channel**: The phase damping channel characterizes the $T2^*$ decay, a loss of quantum coherence, and its Kraus operators are:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\gamma} \end{pmatrix} \quad (5)$$

The parameter γ once more regulates the rate of coherence loss.

- **Bit-flip and Phase-flip channels**: For modeling these channels, the Pauli matrices X (bit-flip) and Z (phase flip) are used on the density matrix ρ . The generic form is:

$$\mathcal{E}_X(\rho) = (1-p)\rho + p(X\rho X^\dagger) \quad (6)$$

$$\mathcal{E}_Z(\rho) = (1-p)\rho + p(Z\rho Z^\dagger) \quad (7)$$

where p is the error probability.

These noise models represent the degradation of quantum states during teleportation, with the parameters p or γ regulating the intensity of the noise.

B. Quantum Teleportation Protocol Simulation

Quantum teleportation is a procedure in which the quantum state of a qubit is transmitted from one party to another through the utilization of entanglement and classical communication. Our simulation adheres to the following general steps:

1) *State Preparation*: A single qubit is developed in a superposition state $|\psi\rangle$, defined by two angles on the Bloch sphere, θ and ϕ . The quantum state can be represented as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (8)$$

2) *Bell Pair Generation*: A maximally entangled Bell state is produced between two parties, Alice and Bob. This entanglement serves as a resource throughout the teleportation process.

3) *Channel Transmission*: The quantum state $|\psi\rangle$ is transmitted through a noisy quantum channel. We use the Kraus operators that were previously presented to model the noise.

4) *Bell Measurement and Classical Communication*: Alice uses her qubit to do a Bell-state measurement and then transmits the result to Bob over a classical communication channel. Bob uses a unitary operation on his qubit, which is determined by the measurement result.

5) *State Reconstruction*: Bob performs an appropriate unitary transformation on his qubit, effectively restoring the teleported state $|\psi\rangle$.

6) *Fidelity Calculation*: The fidelity F of the teleported state in comparison to the original state is determined using the formula:

$$F(\rho, \rho') = \text{Tr}\left(\sqrt{\sqrt{\rho}\rho'\sqrt{\rho}}\right)^2 \quad (9)$$

Where ρ represents the starting point and ρ' denotes the final point after teleportation.

C. Quantum Information Metrics

To assess the quality of the teleportation process under varying noise conditions, we calculate several essential quantum information metrics that offer insights into the system's state.

1) *Fidelity*: The fidelity of a quantum system is defined as its proximity to the ideal state in its ultimate state. This metric is used to evaluate the effectiveness of teleportation and is described as:

$$F(\rho, \rho') = \text{Tr} \left(\sqrt{\sqrt{\rho} \rho' \sqrt{\rho}} \right)^2 \quad (10)$$

2) *Trace Distance*: The trace distance between two quantum states, denoted by ρ and ρ' , can be calculated using the following formula:

$$T(\rho, \rho') = \frac{1}{2} \text{Tr} |\rho - \rho'| \quad (11)$$

This measurement shows how distinguished the two quantum states are from one another.

3) *Von Neumann Entropy*: The Von Neumann entropy $S(\rho)$ quantifies the mixedness of a quantum state:

$$S(\rho) = -\text{Tr}(\rho \log \rho) \quad (12)$$

This is particularly useful when trying to estimate how much data gets lost while teleporting.

4) *Purity*: The purity $\gamma(\rho)$ is defined as:

$$\gamma(\rho) = \text{Tr}(\rho^2) \quad (13)$$

Purity quantifies the degree to which a quantum state is pure, indicating its freedom from noise.

5) *Quantum Fisher Information*: The Quantum Fisher Information (QFI) quantifies how much a quantum state changes in response to a parameter change. The expression is provided by:

$$\mathcal{F}(\rho, \theta) = \sum_{i,j} \frac{2|\langle \psi_i | \frac{\partial \rho}{\partial \theta} | \psi_j \rangle|^2}{\lambda_i + \lambda_j} \quad (14)$$

where λ_i are the eigenvalues of the state ρ , and $\frac{\partial \rho}{\partial \theta}$ is the derivative of the density matrix with respect to the parameter θ .

D. Dataset Overview

We developed a comprehensive dataset by conducting extensive Monte Carlo simulations of QT methods under various noise models. The dataset comprises 15,360 unique instances of quantum channels, with each instance categorized into one of five distinct channel types: depolarizing, amplitude damping, phase damping, bit flip, and phase flip. Each sample is characterized by a 35-dimensional feature vector, which includes physics-informed descriptors obtained from quantum information theory, such as fidelity measures, entanglement metrics, and parameters that can be experimentally adjusted. To achieve a balanced classification task, it is essential that each channel type is represented equally, with a total of 3,072 instances assigned to each type. The parameter space was methodically sampled to encompass a broad spectrum of noise intensities and initial states throughout the Bloch sphere, thus ensuring a thorough representation of realistic variations in quantum systems.

E. Machine Learning for Quantum Channel Classification

We employ ML methods to categorize quantum channels using features retrieved from the teleportation protocol model. These features encompass physical factors such as θ , ϕ , and noise strength, as well as quantum information metrics such as fidelity, trace distance, entropy, and coherence loss.

1) *Feature Selection and Engineering*: *Feature selection* based on *mutual information* and *analysis of variance* (ANOVA) allows us to identify the most distinctive features. These features that were chosen are then standardized using *StandardScaler* to make sure they are uniform and to facilitate simpler ML.

2) *Machine Learning Models*: We employ the following ML models for the classification of quantum channels:

- **Random Forest**: An ensemble learning technique that generates several decision trees and produces the mode of the class labels.
- **Gradient Boosting**: A boosting approach that maximizes classification accuracy by building an additive model of decision trees.
- **Support Vector Machine (SVM)**: An effective supervised ML model for classifying data with a large number of dimensions.
- **Multilayer Perceptron (MLP)**: A neural network-based technique for complicated categorization problems.

These classifiers are evaluated by *stratified sampling* and *cross-validation*, with F1-score and accuracy being the primary metrics. Further, paired t-tests are used for *significance testing* in order to find statistically significant variations in the model performances.

F. Statistical Analysis

To ensure the robustness of the ML models, we perform *statistical analysis* of the cross-validation scores using *confidence intervals* and *t-tests*. This allows us to assess the reliability of the models and make confident comparisons between them.

IV. RESULT ANALYSIS

The following tables provide a detailed performance evaluation of our applied models. The overall performance of SVM, MLP, RF, and GB classifiers is summarized in Table I. To ensure the robustness of the ML models, we perform statistical analysis of the cross-validation scores using confidence intervals and t-tests. This allows us to assess the reliability of the models and make confident comparisons between them. Also, the per-class precision, recall, and F1-scores of the best-performing GB model are highlighted in Table II.

Table I compares the performance of four ML models according to cross-validation and independent test findings. GB had the best overall performance, with the highest cross-validation mean accuracy (0.9615 ± 0.0062), narrowest confidence interval $[0.9568, 0.9662]$, test accuracy of 0.9648, and F1-score of 0.9615, indicating strong generalization ability. Furthermore, RF followed closely with a cross-validation mean of (0.9559 ± 0.0064) and a test accuracy of 0.9577, indicating stable and competitive predictive power. On the other hand,

TABLE I
MODEL PERFORMANCE SUMMARY

Model	CV Mean $\pm$ Std	95% CI	F1-Score	Test Acc.
SVM(RBF Kernel)	0.8900 ± 0.0118	[0.8811, 0.8988]	0.8912	0.8789
MLP	0.9257 ± 0.0082	[0.9195, 0.9318]	0.9258	0.9271
Random Forest	0.9559 ± 0.0064	[0.9510, 0.9607]	0.9558	0.9577
Gradient Boosting	0.9615 ± 0.0062	[0.9568, 0.9662]	0.9615	0.9648

the MLP model fared moderately, with a CV mean of (0.9257 ± 0.0082), test accuracy of 0.9271, and F1-score of 0.9258, whereas the SVM (RBF kernel) did poorly, with a CV mean of (0.8900 ± 0.0118), wider confidence intervals [0.8811, 0.8988], and the lowest test accuracy of 0.8789 and F1-score of 0.8912. Overall, ensemble-based algorithms (GB and RF) beat MLP and SVM, demonstrating their reliability and effectiveness for this task.

TABLE II
PER-CLASS PRECISION, RECALL, AND F1-SCORE OF GRADIENT BOOSTING CLASSIFIER

Class	Precision	Recall	F1-Score
Depolarizing	1.00	1.00	1.00
Bit Flip	0.99	1.00	1.00
Amplitude Damping	0.96	0.92	0.94
Phase Damping	0.94	0.95	0.95
Phase Flip	0.93	0.95	0.94

Table II presents the per-class precision, recall, and F1-score of the GB classifier, indicating its robust and balanced performance across all error classes. The model attained flawless classification for both depolarizing and bit-flip errors, with precision and recall both equal to 1.00, yielding an F1-score of 1.00. The model for amplitude damping achieved a precision of 0.96 and a recall of 0.92, resulting in an F1-score of 0.94. Phase damping and phase-flip errors were classified with high reliability, yielding F1-scores of 0.95 and 0.94, respectively. The results demonstrate the effectiveness of the GB model, consistently achieving high precision and recall across all error types, with notably superior performance in identifying bit-flip and depolarizing errors.

Figure 2 illustrates the significance hierarchy of various attributes utilized by the GB model. The findings reveal that the Pauli-Z change is the most significant factor, succeeded by the Pauli-X change, illustrating that fluctuations in quantum states profoundly influence model predictions. Additionally, the significance of quantum state sensitivity and noise characteristics is reflected in the contributions of features such as anisotropy measure, channel signature, QFI-Z, and trace distance. Conversely, attributes like as reliability and coherence-related metrics exhibit comparatively diminished significance. The research indicates that fluctuations in the Pauli operator and quantum dynamical characteristics are the principal determinants of the model’s predictive efficacy. Moreover, Figure

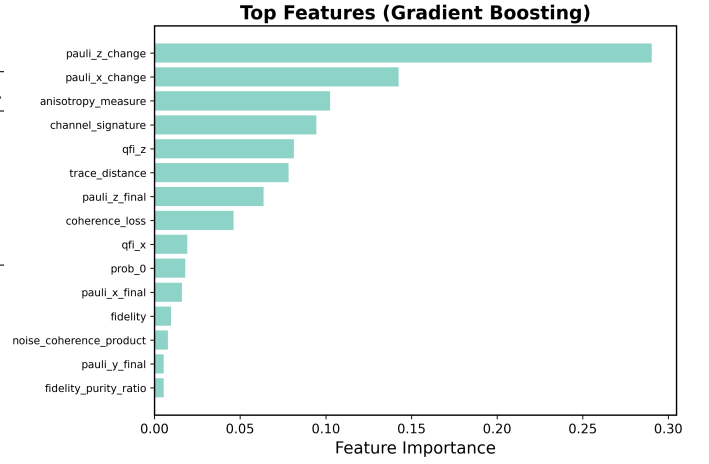


Fig. 2. Gradient Boosting-based important features.

3 displays the learning curves for the GB model, contrasting training and cross-validation performance with an increase in training size. The training accuracy is persistently high (99%), demonstrating effective comprehension of complicated patterns within the dataset. The cross-validation accuracy markedly increases as the sample size grows, advancing from around 70% to almost 94–95%, prior to attaining a plateau. The intersection of training and validation curves implies effective generalization with restricted overfitting. These outcomes affirm the strength of the Gradient Boosting model for anticipating the behavior of quantum systems influenced by noise.

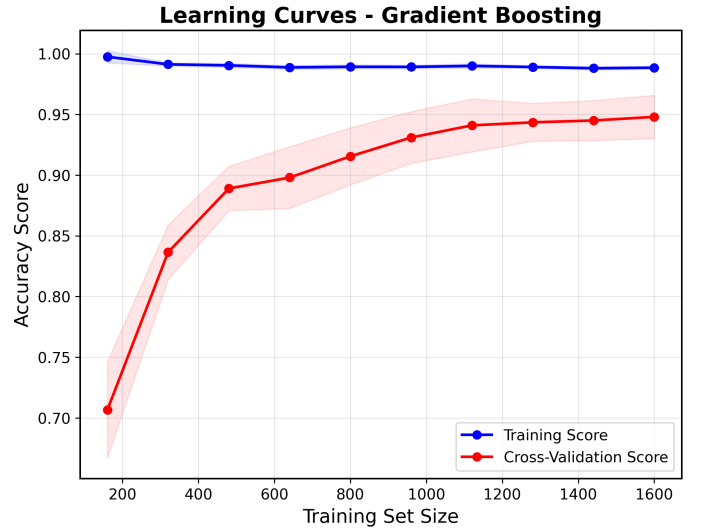


Fig. 3. Learning curve of Gradient Boosting.

Furthermore, Figure 4 illustrates that the mean fidelity diminishes across all noise channels with escalating noise intensity, corroborating the anticipated adverse impact of noise on the preservation of quantum states. Among the inspected channels, phase damping demonstrates the most resilience,

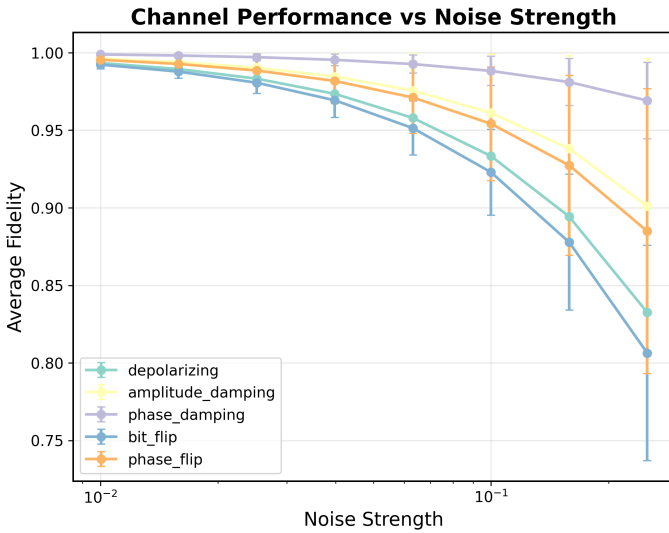


Fig. 4. Impact of noise on channel performance.

preserving fidelity values near one even in the presence of significant noise. While bit flip and depolarizing channels exhibit more noticeable fidelity deterioration at increasing noise levels, amplitude damping likewise exhibits comparatively consistent performance. The significant decrease noted in bit flip noise signifies an increased susceptibility of the system to individual bit errors, whereas the robustness of phase damping implies that phase-related decoherence exerts a relatively minor influence on the fidelity of the examined quantum system.

V. CONCLUSION AND FUTURE WORK

This study developed a physics-informed ML framework for classifying quantum noise channels in teleportation systems. The suggested method adeptly differentiates five representative noise channels by using Kraus-based noise modeling, simulating QT, and extracting quantum information features. The assessment of several classifiers indicates that ensemble approaches, especially GB, attain exceptional classification accuracy of 96.48%, underscoring the efficacy of physics-informed ML in quantum noise characterization. The suggested methodology establishes a basis for creating noise-aware quantum communication protocols by facilitating effective channel identification using quantifiable quantum-state attributes.

Subsequent investigations will concentrate on authenticating the framework with real quantum hardware, integrating hybrid quantum-classical learning paradigms, and formulating adaptive noise reduction techniques for expandable quantum networks.

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